Sparse Arrays for Near-Field Cell-Free XL-MIMO

Crowded vs. Uncrowded Users, the Proposed CAPE Technique,
and How It Differs from Existing Sparse-Array Methods

Explanatory note (simple English)

October 13, 2026

What this document answers, in plain words.

1. Are your 8 users "crowded"? What do crowded and uncrowded mean?
2. Why does crowding matter? Who benefits from the sparse array, and who does not?
3. How are antennas arranged in old arrays (ULA, uniform sparse, ECA), and does the proposed method "stretch" the array in a new way?
4. How is the proposed technique different from the 2025 ECA paper? What exactly is new?
5. The full working of the proposed system, step by step, from the antennas to the user.

1 The short answer first

Your current scenario (8 users spread over the 300×300 m area) is the *uncrowded* case. This is exactly the case where the proposed technique only *ties* the old dense array. The proposed technique shows a clear gain only when the users become *crowded* (many users close together in a small spot).

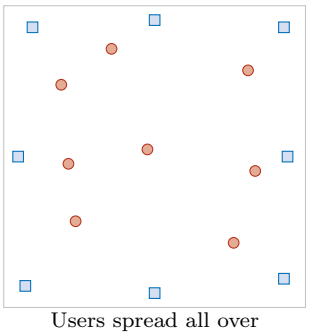
Two more things to fix in your mental picture before the details:

- **The proposed method does *not* invent a new antenna layout.** It uses the same ECA (extended co-prime array) layout from the 2025 paper. We do *not* "stretch the array in a new way".
- **What is new is the *signal processing*, not the array.** The new part is how several access points (APs) *cooperate* to estimate the channel of a sparse array. We call it **CAPE** (Cross-AP Position-fused Estimation). This is the honest novelty, and it is what makes the sparse array actually useful in a cell-free network.

2 What "crowded" and "uncrowded" users mean

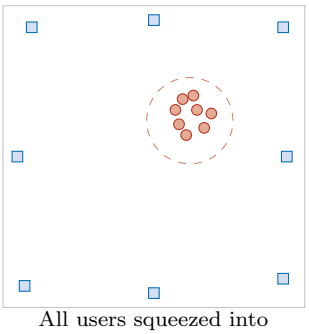
Picture the service area as a 300 m × 300 m square. There are $L = 8$ access points (APs) placed around it, and $K = 8$ users somewhere inside it.

Uncrowded (your baseline)



Users spread all over the area. Each user is close to a *different* AP.

Crowded (hotspot)



All users squeezed into one small spot (a 25 m circle), near the *same* AP.

Uncrowded ("baseline" in the code). Users are spread across the whole area. In the code each user is dropped around a different AP, at a distance of 20–150 m. Because they sit in different directions and different distances, *the network can already tell them apart easily.* **This is your current 8-user setup.**

Crowded ("hotspot" in the code). All users sit inside one small circle (radius 25 m by default; the sweep also tries 10 m, which is very crowded). A real example: everyone in one lecture hall, one stadium stand, one train platform, one conference room. To a nearby AP, all these users arrive from nearly the *same direction* and nearly the *same distance*, so *their radio channels look almost identical.*

The one-line definition. Users are *crowded* when their channels look similar to an AP (same direction, similar distance), so the AP struggles to separate them. They are *uncrowded* when their channels look different, so separation is easy.

3 Why crowding decides who benefits

A big antenna array is a tool for *separating* users in space. If the users are already easy to separate, a better tool does not help. If the users are hard to separate, a better tool helps a lot. That is the whole story.

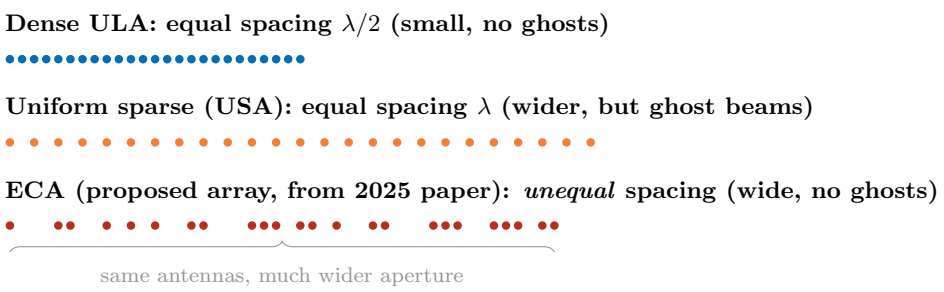
	Uncrowded users	Crowded users
Do their channels look alike?	No. Different directions / distances.	Yes. Nearly the same.
Can the network separate them already?	Yes, easily (many APs see them differently).	No, this is the hard part.
Does extra array resolution help?	**Almost not at all.** The job is already done.	**Yes, a lot.** This is the bottleneck.
So the sparse array...	... gives no real gain (a tie).	... gives a clear gain.

Why "should" users be crowded for the gain to appear? It is not that crowding is good for the users — crowding is actually a hard, stressful situation for the system (lots of mutual interference). It is that *crowding is the situation where a better array earns its keep.* A sharper tool only shows its value on a hard job. When the job is easy (uncrowded), any tool looks the same, so the sparse array and the dense array tie. When the job is hard (crowded), the sparse array's extra resolution finally matters, and the proposed technique wins.

This is exactly what your results showed (the crowding-sweep figure): as the hotspot radius shrinks from 200 m down to 10 m, the proposed curve pulls further and further below the dense array. At 10 m the proposed technique had about 2.5–4× lower bit-error rate; at 100 m they met.

4 How antennas are arranged: dense, uniform-sparse, ECA

All of these use the *same number* of antennas ($N \approx 64$). The only difference is *where* the antennas sit. "Stretching" the array means spreading the same antennas over a wider line, which gives a bigger aperture and sharper resolution — but spreading them wrongly creates ghost beams (grating lobes).



Array	Spacing	Aperture	Ghost beams	Note
ULA (dense)	equal, $\lambda/2$	small (3.15 m)	none	poor range resolution
USA (uniform sparse)	equal, λ	wide (6.3 m)	**strong**	ghosts cause confusion
ECA (non-uniform)	**unequal**	widest (7.6 m)	none (suppressed)	weak scattered sidelobes

ULA (old, dense). You "stretch" nothing: antennas are packed at half a wavelength. The array is short, so it cannot focus in *distance* well; two users on the same line (one near, one far) look almost identical.

USA (old, uniform sparse). You "stretch" the array *uniformly*: keep equal spacing but make it larger. The array is now wide and sharp, *but* equal spacing creates exact copies of the beam at other angles — *grating lobes* (ghost beams). A user sitting on a ghost beam is confused with the real one.

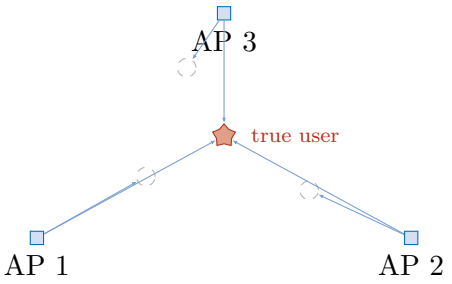
ECA (the array we use, from the 2025 paper). You "stretch" the array *non-uniformly*: antennas sit at *unequal* gaps following a co-prime pattern. Because the gaps are irregular, the ghost copies never line up, so there are no strong ghost beams. The price is a floor of weak scattered sidelobes.

So does the proposed method stretch the array? It uses the ECA stretch — non-uniform, wide, no strong ghosts. **We do not change how a *single* array is stretched.** Our new idea works at the *network* level, across many APs, and is described next.

5 What is actually new: the CAPE technique

A sparse array has one weakness: it is *ambiguous*. Its beam is very sharp, but it also has several "maybe" spots where a user could be. One AP alone cannot always tell the true spot from the false ones — this is what hurt the plain ECA and made its channel estimates worse than the dense ULA.

Key idea of CAPE (one sentence). A user has only *one* true physical location, and every AP's channel is a known function of that location; so if we *combine* the sparse arrays of several APs, the true location is the only spot they all agree on — the false spots of one AP are killed by the others.



Each AP sees the *true* spot plus its own *false* spots (grey).

Only the true spot is pointed at by *all* APs. CAPE keeps that one and throws the rest away.

How CAPE differs from the 2025 ECA paper

	2025 ECA paper (Zhou *et al.*)	**This work (proposed)**
Array geometry	ECA (they proposed it)	**Same ECA (we reuse it, we cite them)**
How many arrays	**One** array / base station	**Many** APs (cell-free, $L = 8$)
Direction	Downlink (transmit beamforming)	**Uplink** (receive / data detection)
Main contribution	Beam-pattern analysis, hybrid beamforming for one array	**CAPE: APs cooperate to remove the sparse array's ambiguity in channel *estimation***
Ambiguity handling	Analysed for a single array	**Removed by fusing several APs**
Detector	—	Cross-AP list SIC (your earlier work)

Be honest in the paper. Do *not* claim the ECA array as your invention — it is from the 2025 paper, and the co-prime idea is older still (Vaidyanathan & Pal, 2011). What you *can* claim as new is: (i) putting sparse ECA arrays at *every* cell-free AP; (ii) the **CAPE** cross-AP position-fused channel estimation that resolves the sparse-array ambiguity; and (iii) pairing it with your cross-AP list SIC detector. Search the 2025–2026 literature once more before writing the word "first".

6 Full working, step by step (antenna to user)

Setup. There is no single big base station. Instead there are $L = 8$ small APs spread around the 300 m square, all connected to one central processor (CPU). *Each AP carries one sparse ECA array of 63 antennas, 7.6 m wide.*

Step 0 — How we make one array sparse (the procedure)

1. Start from the half-wavelength grid (positions $0, \frac{\lambda}{2}, \lambda, \dots$).
2. Pick two co-prime numbers, $M = 5$ and $N = 4$ (no common factor).
3. Place one small "block": 5 antennas spaced $4\times$ apart (at $0, 4, 8, 12, 16$) plus 4 antennas spaced $5\times$ apart (at $0, 5, 10, 15$), sharing the first antenna. That is 8 antennas at unequal gaps.
4. Repeat this block 8 times with a fixed period, mirror it about the centre. Result: 63 antennas over 7.6 m, with *unequal* spacing — the ECA.

Every antenna is an ordinary receive antenna; we did not change the hardware, only *where* the antennas are mounted on the panel. Fewer-but-wider spacing = larger aperture with the same number of radio chains.

Step 1 — Pilots (uplink)

Each user sends a short known "pilot" signal. Every AP's 63 antennas receive it. Because the array is sparse, each AP's own guess of "where is this user" has one sharp true peak *and* a few false peaks (the ambiguity).

Step 2 — CAPE: the APs vote on the true position

1. The 3 strongest APs each send the CPU a few candidate positions (their top peaks).
2. The CPU takes each candidate and checks it against the received pilots of the 4 strongest APs together (a joint score). The true position scores high at *all* APs; a false peak of one AP scores low at the others, so it loses.
3. The CPU keeps the best position and polishes it with a small 2-D search.

Step 3 — Each AP builds its channel from the agreed position

The CPU sends the one agreed position back. Every AP now knows exactly where the user is, so it builds its own channel (its own set of 63 complex numbers) from that position, and fine-tunes it locally. A safety "gate" keeps an AP's own local estimate instead, whenever that local estimate is clearly better (this is why the method is never worse than plain local estimation at high signal strength).

Step 4 — Data detection

With good channels in hand, the users' data symbols are detected using the cross-AP list SIC detector from your earlier work: APs cancel already-decoded users and, when unsure, fuse all APs' observations to resolve the hard cases.

Why this helps. The sparse array gives *sharp* beams (good), but *ambiguous* ones (bad). CAPE removes the "bad" part by making the APs agree on one position, and keeps the "good" part (sharpness). So we finally get the sparse array's resolution *without* paying its estimation penalty.

7 What each type of user gains

Crowded users — the big winners. When many users sit close together, the dense ULA cannot tell them apart and its interference is high. The wide ECA separates them in both *angle and distance*; CAPE makes the channel estimation reliable; and the cross-AP detector finishes the

job. Result (from your runs): at a 10 m hotspot the proposed system had about **2.5–4× lower bit-error rate** and clearly higher sum-rate than the dense ULA, plus it located each user to within a few centimetres.

Uncrowded users — roughly a tie. When users are spread out, the network separates them easily already, so the extra resolution is not needed. Here the proposed system matches the dense ULA (within a couple of percent, sometimes a hair worse), and it costs more pilot overhead. **For purely uncrowded deployments the sparse array is not worth it** — that is the honest conclusion, and it is why your 8-user spread-out run did not show a gain.

8 Summary of the measured results (from your 10 figures)

What was measured	Uncrowded (baseline)	Crowded (hotspot)
Channel-estimation error (NMSE)	CAPE best; ECA+local worst	CAPE fixes ECA (big gain)
BER, final detector (List+CrossAP)	tie (within noise)	proposed best, up to 4× lower
Sum-rate	≈+1.7%	≈+4.4% (more at 10 m)
Cell-edge (5% outage) SE	CAPE > local	ECA/ULA close
User localisation (bonus)	ECA ∼20× sharper	ECA ∼20× sharper
Data fronthaul (gate rate)	slightly lower	**clearly lower**
Extra pilot fronthaul (cost)	+256 scalars / user / block	+256 scalars / user / block
Estimation run time (cost)	∼1.5× slower	∼1.5× slower
CAPE on a *dense* ULA	bad (do not do it)	bad (do not do it)

Two cautions on these numbers. (1) They come from only 12 setups, so small wiggles (especially the very-crowded 10 m point) may be sampling noise — re-run with 30–50 setups before quoting exact values. (2) The "likelihood maps" figure (PROP-1) is not reliable yet because of a plotting bug (only the x-coordinate was used); it will be fixed separately and does not affect any of the number-based figures.

9 How to position this in a paper (one paragraph)

Frame it as a *crowded-scenario, cooperation-aided* sparse-array method, not a universal win. The message: "In near-field cell-free XL-MIMO, a non-uniform sparse (ECA) array per AP raises spatial resolution but, used alone, is hard to estimate and gives no network gain. We propose CAPE, a cross-AP position-fused channel estimator that resolves the sparse-array ambiguity by making APs agree on one physical position, and we combine it with cross-AP list SIC. CAPE restores and exceeds dense-array performance when users are spatially crowded (hotspots), ties it when users are spread out, lowers data fronthaul, and provides centimetre-level user localisation as a by-product — at the cost of extra pilot overhead." Lead with the crowding-sweep and localisation figures; be upfront about the pilot cost and the uncrowded tie.